

Machine Learning-Enhanced Performance Analysis of an IoT-Enabled Solar EV Charging Station

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Abstract—In this study, we present an IoT-enabled, small-scale solar-powered EV charging station with real-time monitoring and control. The system allows remote data collection, which is utilized to estimate the maximum power point (MPP) without requiring a dedicated MPPT unit, replaced by a pulse width modulation (PWM) charge controller, and to predict power generation useful for fault detection and energy planning. Single-axis tracking, driven by an ESP32, light-dependent resistors (LDRs), and a servo motor, maximizes solar capture. Real-time voltage and current data are displayed via Wi-Fi on an OLED and a web interface. Experimental results show that with a 27 W input and a 25 W output, the system achieves an efficiency of 92.6%. Collected current–voltage–temperature data estimate the photovoltaic I–V curve with a 2.5% error in operating current. A machine learning framework predicts solar power output from six environmental features—irradiance, temperature, dew point, humidity, wind speed, and sky condition—using ridge regression ($R^2 = 88\%$). After feature selection, a support vector regression model achieves $R^2 = 85.3\%$ using only irradiance and sky condition. Compact, modular, and low-cost, the proposed design offers a practical model for sustainable EV charging integrating renewable energy, IoT, and data-driven intelligence.

Index Terms—Electric vehicle charging, IoT, PWM charge controller, Irradiance, Solar tracking, Machine learning

I. INTRODUCTION

Road transport remains the dominant mode of commuting worldwide due to its affordability, accessibility, and capacity to support large-scale passenger and freight movement. However, it faces growing criticism for its dependence on fossil fuels and the associated environmental impacts, including greenhouse gas emissions, air pollution, and noise [1]. Conventional internal combustion engine vehicles (ICEVs) are also inefficient, converting only about 15% of fuel energy into propulsion, which, alongside concerns over energy security and climate change, has accelerated the shift toward electric mobility.

Electric vehicles (EVs) have seen remarkable adoption across diverse transport applications, from two-wheelers to buses and heavy-duty trucks. Declining battery costs, higher energy densities, and advances in power electronics have enhanced their competitiveness. Plug-in electric vehicles (PEVs) can convert up to 75% of energy into propulsion while producing zero tailpipe emissions, with one gallon of gasoline equivalent to roughly 33.7 kWh of electricity [2].

At the same time, photovoltaic (PV) technology has become increasingly cost-effective and scalable. Integrating EV charging with PV systems enables direct utilization of solar energy while allowing EV batteries to serve as distributed storage, thereby mitigating intermittency, easing grid demand, and improving grid stability under adverse weather conditions [3], [4]. Due to their limited storage capacity and shorter travel range, electric vehicles require charging stations to supply the energy needed for long-distance travel. Charging methods are generally categorized into AC and DC modes. Advances in energy conversion technology, battery management systems, installation methods, and design standards have greatly enhanced the feasibility of using photovoltaic systems for electric vehicle charging. Given that EVs are often parked during daylight hours, solar-powered “charging-while-parking” systems offer lower operational costs, reduced CO₂ emissions, and enhanced energy independence. Additionally, PV-integrated charging stations can provide shelter and encourage adoption of clean mobility infrastructure [5].

To enhance the efficiency and usability of solar powered EV charging stations, IoT technology has been increasingly integrated for real time monitoring and intelligent control. IoT enabled systems allow users to remotely access key parameters such as voltage, current, and energy consumption through web or mobile dashboards, improving transparency and operational management. Beyond monitoring, automatic control of solar tracking systems can be implemented via microcontrollers interfaced with light dependent resistors (LDRs) or other sensors, enabling the panels to follow the sun’s trajectory throughout the day and maximize energy capture [6].

Despite these advancements, existing solutions often remain limited in scalability, cost-effectiveness, and the seamless integration of monitoring, control, and hybrid power management. Many prior works focus either on IoT-based monitoring or solar tracking automation in isolation, without combining them into a unified, low-cost, and user-friendly framework that ensures continuous EV charging even under variable solar irradiance conditions. Furthermore, most existing PV-based charging systems depend on dedicated maximum power point tracking (MPPT) controllers, which increase cost and system complexity. However, real-time current–voltage–temperature (I–V–T) data collected from the station can be used to estimate

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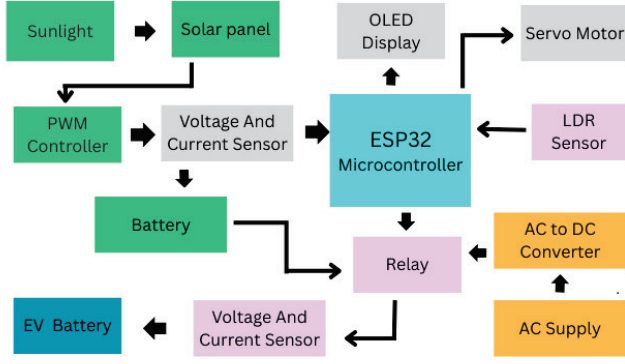


Fig. 1: Block diagram of the proposed EV charging station.

the I-V curve and determine the maximum power point (MPP) computationally, thereby simplifying control without requiring specialized hardware. Additionally, limited research has explored the use of ML techniques, such as MLP [7], [8], ANN [9], [10], RF [11], [12], and polynomial regression [13], for predicting power generation in small-scale PV-powered EV charging systems, despite their potential to forecast energy availability, detect faults, and support adaptive energy management.

In this paper, we present the design and implementation of a solar-powered EV charging station that integrates IoT-based monitoring, pulse-width modulation (PWM)-based charge control, and automatic solar tracking for maximizing irradiance capture throughout the day. The proposed system estimates the MPP from measured voltage, current, and temperature data, eliminating the need for conventional MPPT hardware. Moreover, an ML framework is developed to predict the power output based on six environmental parameters—irradiance, temperature, dew point, humidity, wind speed, and sky condition—using four distinct ML models. Selection of features and principal components (PCs) is further investigated by evaluating all possible feature and PC combinations to identify the most influential predictors. The proposed solution aims to deliver a reliable, cost-effective, and environmentally sustainable approach for next-generation EV charging infrastructures, bridging the gap between clean energy generation, intelligent control, and data-driven power prediction.

II. METHODOLOGY

A. System Overview and Power Conversion

A monocrystalline PV panel converts incident solar irradiance into direct current (DC) power, expressed as

$$P_{solar} = G \times A \times \eta_{pv}, \quad (1)$$

where G is the solar irradiance (W/m^2), A is the panel area, and η_{pv} is the photovoltaic efficiency. To enhance energy capture, a single-axis solar tracker is employed. Two orthogonally positioned LDR sensors detect differential irradiance, and the

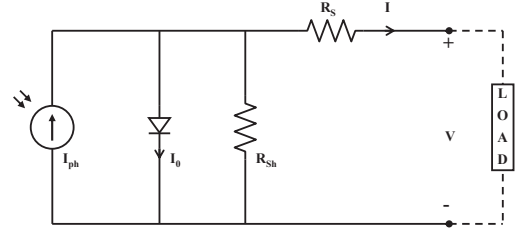


Fig. 2: Equivalent single-diode circuit model of a solar cell

ESP32 microcontroller drives a servo motor to orient the panel for maximum exposure.

The harvested DC power is regulated through a PWM charge controller, where

$$P_{in} = V_{in}I_{in}, \quad P_{out} = V_{out}I_{out}. \quad (2)$$

The controller efficiency is defined as

$$\eta = \frac{P_{out}}{P_{in}} \times 100\%. \quad (3)$$

Voltage is sensed using a divider network,

$$V_{sensed} = V_{in} \frac{R_2}{R_1 + R_2}, \quad (4)$$

and current is measured by an ACS712 Hall-effect sensor:

$$I = \frac{V_{ACS} - V_{offset}}{S}, \quad (5)$$

where V_{offset} is the quiescent voltage and S is the sensitivity. These allow real-time monitoring of power flow and efficiency.

B. Energy Management and IoT Control

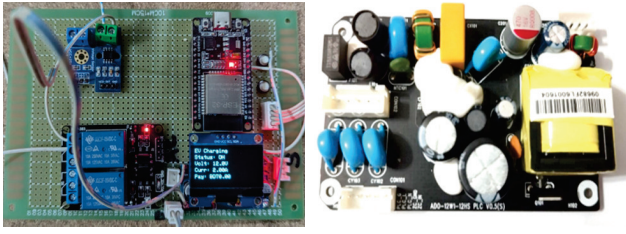
The regulated output charges a 12 V lithium-ion battery bank monitored by a Battery Management System (BMS) to prevent overcharge, over-discharge, and short-circuit faults. The stored energy is computed as

$$E_{stored}(t) = \int_0^t (P_{out} - P_{EV} - P_{loss}) dt, \quad (6)$$

where P_{EV} is the charging power and P_{loss} denotes system losses, ensuring stable operation under varying irradiance.

The ESP32 microcontroller serves as the central unit, acquiring voltage and current data, executing tracking and control algorithms, and handling relay switching. It transmits real-time data to a cloud server via Wi-Fi for IoT monitoring, while an OLED screen locally displays parameters such as battery state-of-charge, charging power, and cost.

To ensure uninterrupted operation, two relays manage hybrid power flow. When solar or battery energy is adequate, the first relay delivers P_{out} to the EV; otherwise, the second relay automatically connects to the grid. This hybrid configuration guarantees continuous and reliable charging.



(a) Main control board (b) Adapter for auxiliary supply

Fig. 3: Circuit design of the proposed EV charging station.

C. Solar Cell Modeling and Parameter Extraction

The solar cell is modeled using the widely accepted *single-diode model*, represented by the equivalent circuit shown in Fig. 2. The circuit consists of a photocurrent source (I_{ph}), a diode with reverse saturation current (I_0) and ideality factor (n), a series resistance (R_s), and a shunt resistance (R_{sh}). The current-voltage relationship is described by

$$I = I_{ph} - I_0 \left[\exp \left(\frac{V + IR_s}{nV_t} \right) - 1 \right] - \frac{V + IR_s}{R_{sh}}, \quad (7)$$

where $V_t = kT/q$ is the thermal voltage, k is the Boltzmann constant, q is the electron charge, and T is the cell temperature in Kelvin.

The experimental dataset consists of measured voltage (V), current (I_{meas}), and temperature (T). The temperature is converted from Fahrenheit to Kelvin as:

$$T_K = \frac{5}{9} (T_F - 32) + 273.15. \quad (8)$$

The parameters I_0 , n , R_s , R_{sh} , and I_{ph} are extracted by fitting the diode model to the measured I - V data. Since Eq. (7) is implicit in I , the current at each voltage is computed numerically using a root-finding method:

$$f(I) = I_{ph} - I_0 \left(e^{\frac{V + IR_s}{nV_t}} - 1 \right) - \frac{V + IR_s}{R_{sh}} - I = 0. \quad (9)$$

The residuals between measured and modeled current are defined as:

$$\text{residuals} = I_{meas} - I_{model}. \quad (10)$$

Optimal parameters are obtained by minimizing the sum of squared residuals using the least-squares method:

$$\min_{I_0, n, R_s, R_{sh}, I_{ph}} \sum_{i=1}^N (I_{meas,i} - I_{model,i})^2, \quad (11)$$

where N is the total number of measured points.

III. CIRCUIT DESIGN

The hardware implementation of the proposed EV charging station is divided into two major functional modules: the main control board and the adapter board, as shown in Fig. 3 (a) and (b) respectively.

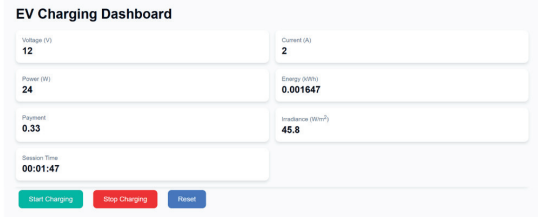


Fig. 4: IoT-based administrative dashboard interface

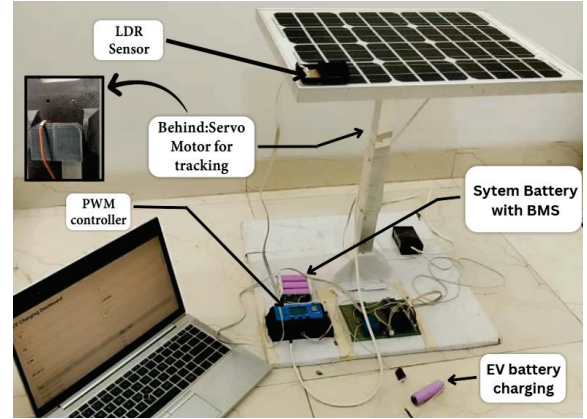


Fig. 5: Full setup of the proposed IoT-enabled system

A. Main Control Board

The main control board manages energy regulation, monitoring, and control. It channels PV panel energy through a PWM charge controller to the battery, preventing overcharging and deep discharging. While PWM is used here for simplicity and cost-effectiveness, the design allows future upgrade to MPPT for higher efficiency. At its center, the ESP32 microcontroller executes charging algorithms, processes sensor feedback from voltage and current sensors, and controls relay activation to route power as needed. Integrated protection circuits safeguard against short circuits, reverse polarity, and battery faults. By combining regulation, sensing, protection, and communication, the main board ensures safe, reliable, and intelligent energy management.

B. Adapter Circuit Board

The adapter board provides a stable auxiliary supply for the ESP32, OLED, and sensors, using DC-DC converters and noise isolation to ensure accurate readings and reliable operation. By separating low-power circuits from the main channel, it enhances robustness and maintains uninterrupted control under fluctuating inputs. Together with the main control board, it manages energy harvesting, storage, and delivery efficiently, while the modular design allows future upgrades such as larger solar arrays, bidirectional charging with advanced IoT-based energy management.

IV. RESULTS AND DISCUSSION

The proposed IoT-enabled solar-powered EV charging station was tested under real-world conditions to assess its

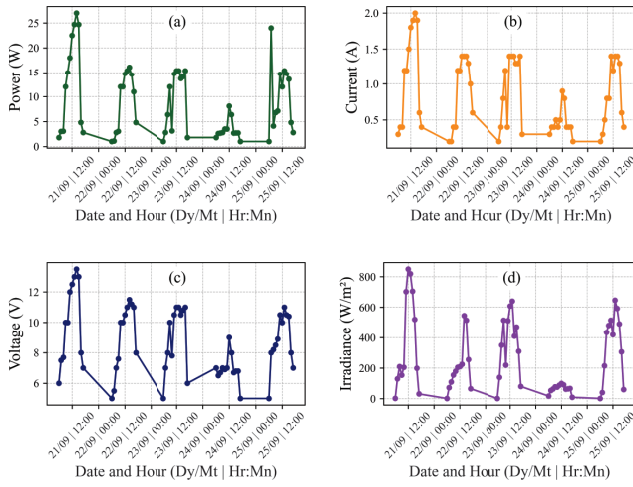


Fig. 6: Hourly variation of (a) power, (b) current, (c) voltage, and (d) irradiance over five days.

TABLE I: Cost analysis of the proposed charging station.

Component	Quantity	Cost/unit (\$)	Total cost (\$)
PV panel	1	500	500
PWM controller	1	18	18
Current sensor	2	4.5	9
Voltage sensor	2	3.5	7
LDR sensor	2	1.25	2.5
Two-channel relay module	1	2.75	2.75
1.3" OLED display	1	8	8
Solar mount and supports	—	35	35
Servo motor (MG995)	1	6	6
Adapter circuit	1	18	18
Buck module	1	1.5	1.5
Cables	—	7	7
ESP32 microcontroller	1	4	4
JST connector, PCB board	—	10	10
Total	—	—	628.75

performance, usability, and cost-effectiveness. An IoT-based dashboard (Fig. 4) provides remote monitoring and control, displaying real-time charging activity, voltage, current, and cost data for operational transparency. Additionally, an on-device OLED screen shows live system parameters, enabling on-site users to monitor energy consumption and battery status without external devices.

A. System Setup and Operational Performance

The complete 30 W experimental setup is shown in Fig. 5. The compact hardware integrates solar harvesting, battery storage, grid backup, and IoT monitoring into a modular unit suitable for both urban and rural deployment. To evaluate system performance, voltage, current, power, and irradiance were recorded hourly from 6:00 AM to 5:00 PM over five consecutive days. Results in Fig. 6 show a clear correlation between solar irradiance and system output, with voltage, current, and power peaking around midday under high irradiance and decreasing during early morning and late evening. This consistent trend confirms the strong dependence of output on solar availability and highlights the effectiveness of the single-axis solar tracking system in maximizing energy capture.

TABLE II: Comparison of the proposed system with existing EV charging solutions

Parameter	Existing Work	Proposed System	Remarks
Gain of Solar Tracking	Increase in efficiency [14]	Increase in energy efficiency	More reliable
Monitoring System	IoT-based energy monitoring with ESP32 [15]	Integrated IoT dashboard + OLED + sensors	More comprehensive
Battery Utilization	Lithium-ion battery [16]	Lithium-ion with BMS control	Improved reliability
Backup Supply	Battery only [17]	Battery + partial grid support	More integrated

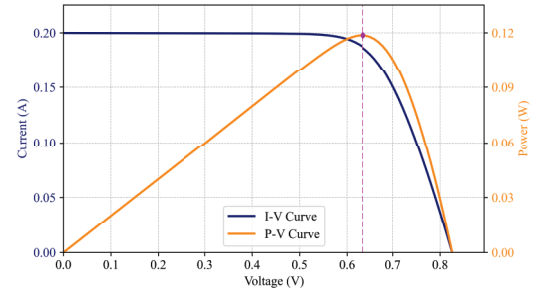


Fig. 7: Estimated current-voltage and power-voltage curves

B. Efficiency, Cost and Comparative Analysis

On September 21, 2025 at 13:00, the recorded solar irradiance was about 820 W/m^2 , yielding a panel output of 24.7 W at 13 V and 1.9 A . The effective surface area of the solar panel was 0.175 m^2 ($50 \text{ cm} \times 35 \text{ cm}$). Considering the expected efficiency of a commercial monocrystalline silicon panel ($\eta_{PV} = 18.5\%$), the theoretical output power from Eq. (1) is $820 \times 0.175 \times 0.185 = 26.55 \text{ W}$. Thus, the system efficiency is $(24.7/26.55) \times 100\% \approx 93\%$ from Eq. (3). Averaging across all samples yields an overall system efficiency of 92.6% . These results indicate that the integration of the PWM controller and solar tracking effectively minimized energy losses and ensured stable power distribution under peak irradiance conditions.

Although the framework was verified with a 30 W solar panel, the system was architecturally designed to accommodate up to 1 kW of solar input. Such scalability allows the design to be extended from small demonstration setups to higher-power EV charging applications without major structural modifications. A detailed component cost analysis was performed for a 1 kW system, showing a total cost of $\$628.75$, as given in Table I. This demonstrates that the station remains low-cost and accessible while ensuring the reliability and performance required for practical EV charging.

To highlight the advantages over existing solutions, Table II summarizes key performance parameters. The proposed system achieves higher energy efficiency through single-axis solar tracking. The IoT monitoring framework, combining a dashboard, OLED display, and multiple sensors, offers a more comprehensive and user-friendly interface compared to prior implementations. The integration of a Battery Management

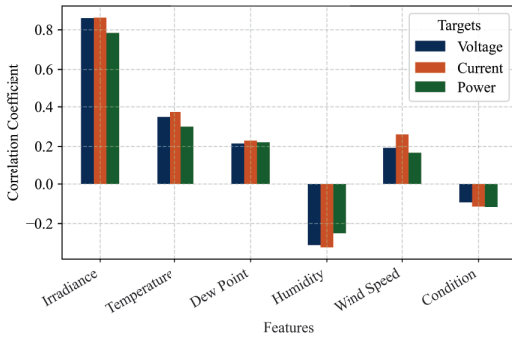


Fig. 8: Correlation of power output with features

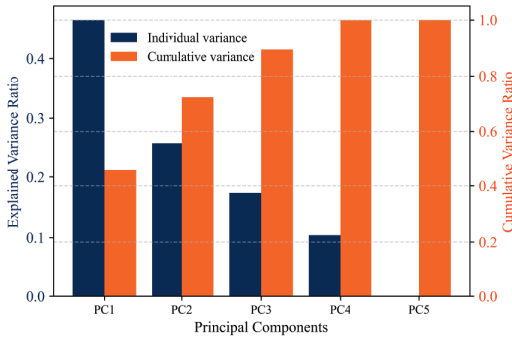


Fig. 9: Explained and cumulative variance ratios of PCs

System ensures reliable battery operation, while the hybrid supply strategy with partial grid support guarantees continuous operation under low irradiance. Collectively, these features make the station more versatile, practical, and reliable for real-world deployment.

C. Solar Cell Characteristic Curves

The single-diode model was used to generate the current–voltage (I–V) and power–voltage (P–V) characteristics from experimental data. As shown in Fig. 7, the I–V curve (blue) follows typical solar cell behavior, starting at the short-circuit current ($I_{sc} \approx 0.2$ A) and falling to zero at the open-circuit voltage ($V_{oc} \approx 0.83$ V). The P–V curve (orange) peaks at the maximum power point (MPP), where $V_{mpp} \approx 0.635$ V, $I_{mpp} \approx 0.187$ A, and $P_{mpp} \approx 0.119$ W. Using the proposed approach, the corresponding single-diode cell parameters were determined as $I_0 = 5 \times 10^{-12}$ A, $R_s = 0.51589$ Ω , $R_{sh} = 999.99996$ Ω , and ideality factor $n = 1.29986$. At the operating voltage of 0.62 V, the model estimates a current of 0.195 A, whereas the experimentally measured current is 0.20 A, resulting in a relative deviation of approximately 2.5

D. Correlation and Principal Component Analysis

In addition to irradiance, the dataset from [18] contains five other environmental variables: temperature, dew point, humidity, wind speed, and sky condition. Fig. 8 shows the correlation between environmental variables and photovoltaic

TABLE III: Performance of ML models to predict PV power

Model	R ²	RMSE	MAE	R
Ridge	0.880	0.282	0.239	0.957
Gradient Boosting	0.481	0.515	0.382	0.774
SVR	0.449	0.557	0.463	0.724
KNN	0.373	0.646	0.584	0.693

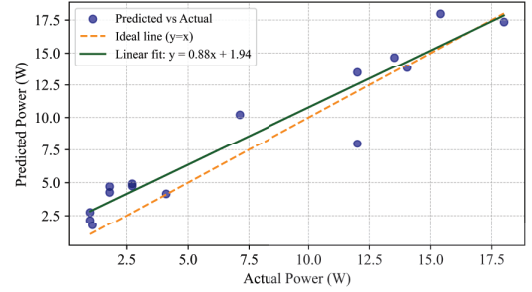


Fig. 10: Predicted versus actual PV module power outputs.

(PV) outputs. Irradiance exhibits the strongest positive correlation ($r > 0.8$) with voltage, current, and power, confirming its dominant influence on PV performance. Temperature has a moderate positive correlation (0.3–0.37), while dew point and wind speed show weak positive correlations (≈ 0.2). Humidity and weather condition show weak negative correlations (≈ -0.25 to -0.1) due to reduced irradiance under cloudy or humid conditions.

Principal Component Analysis (PCA) was performed to identify the most influential environmental features affecting photovoltaic (PV) power output and to reduce data dimensionality. Fig. 9 presents the explained variance ratio (blue bars) and cumulative variance ratio (orange line) for the six principal components derived from the dataset.

The first three components (PC1–PC3) account for roughly 90% of the total variance, indicating that irradiance, temperature, humidity, wind speed, dew point, and sky condition can be effectively represented by these components. PC1 explains about 44% of the variance, while PC2 and PC3 contribute approximately 23% and 17%. The remaining components (PC4–PC6) contribute less than 10%, indicating limited additional information.

E. Comparison of Regression Model Performance

As shown in Table III, Ridge regression achieved the highest accuracy ($R^2 = 0.88$, $R = 0.957$), outperforming Gradient Boosting, SVR, and KNN models. This indicates that linear regularized regression effectively captures the relationship between environmental features and PV power output for this dataset.

Fig. 10 compares predicted and measured PV power values. The regression line $y = 0.88x + 1.94$ shows a strong linear trend, with most data points lying close to the ideal $y = x$ line, confirming the reliability of the Ridge regression model for power estimation.

TABLE IV: Top feature (fts) subsets and top principal component (PC) subsets with model performance.

Model	Top Features	Top PCs	R ² (fts)	R ² (PCs)
Ridge	Irradiance, Temperature, Humidity, Wind Speed, Condition	PC1, PC2, PC3, PC4	0.881	0.691
SVR	Irradiance, Condition	PC1, PC2, PC3	0.853	0.499
KNN	Irradiance, Dew Point	PC1, PC2, PC3	0.784	0.456
GB	Irradiance, Temperature, Condition	PC2, PC3, PC4, PC5	0.779	0.166

F. Impact of Feature and Principal Component Selection

Feature selection results highlight **irradiance** as the most influential variable across all models, confirming its dominant effect on PV power generation. Ridge regression achieves the highest accuracy ($R^2 = 0.881$) using a broader feature set, while SVR attains comparable accuracy ($R^2 = 0.853$) with only irradiance and condition, demonstrating that nonlinear models can generalize effectively with fewer but more informative features. KNN and Gradient Boosting (GB) show reduced accuracy, indicating sensitivity to irrelevant or redundant variables. In contrast, when models are trained on the top principal components (PCs), performance generally decreases. As shown in Table IV, Ridge regression achieves $R^2 = 0.691$ using the first four PCs, while SVR and KNN record lower values around 0.5. GB exhibits the poorest performance ($R^2 = 0.166$), reflecting its reliance on direct feature relationships rather than transformed components.

The experimental evaluation confirms that the IoT-enabled solar EV charging station efficiently harvests solar energy and manages power flow through hybrid switching and real time monitoring. The test solar cell exhibited $V_{oc} = 0.83$ V, $I_{sc} = 0.2$ A, and an MPP efficiency of 14%. Machine learning models, particularly Ridge and SVR, demonstrated strong predictive capability for PV power based on environmental data, supporting intelligent monitoring and planning. Future work will focus on integrating AI-driven fault detection, adaptive control, and predictive energy management to enhance scalability and resilience in sustainable EV charging infrastructure.

V. CONCLUSION

This paper presents a low-cost, solar-driven, IoT-based EV charging station for sustainable infrastructure. The system employs a 30 W monocrystalline solar panel characterized by $V_{oc} = 0.83$ V, $I_{sc} = 0.2$ A, and a maximum power point at $V_{mpp} = 0.635$ V and $I_{mpp} = 0.187$ A. A PWM charge controller enables efficient energy harvesting, while a 12 V lithium-ion battery with an integrated BMS ensures safe charging and discharging. Charging control is managed via a two-channel relay that automatically switches between solar-battery and grid power. Single-axis solar tracking using an LDR sensor and ESP32 microcontroller maximizes irradiance capture, achieving an overall system efficiency of about 92.6%. IoT monitoring provides real-time voltage, current, and power data through both an OLED display and an online dashboard, with partial grid support maintaining operation under low

irradiance. Machine learning analysis using environmental features—irradiance, temperature, humidity, wind speed, and condition—demonstrated strong predictive performance for PV power output, with Ridge regression achieving $R^2 = 0.88$ and SVR $R^2 = 0.85$. The prototype exhibits stable performance, modular scalability, and effective PWM-based battery protection, offering a practical and replicable model for decentralized, renewable-powered EV charging with data-driven intelligence.

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